

EDA

Meghan Duarte

2026-05-23

Clean the Data

```
library(dplyr)
library(tidyverse)
library(ggplot2)
library(tsibble)
library(feasts)
library(fable)
```

```
singapore <- read.csv('BirthsAndFertilityRatesAnnual.csv')
```

```
clean_1 <- singapore |>
  filter(DataSeries %in% c("Total Fertility Rate (TFR)", "Total Live-Births"))
```

```
clean_2 <- pivot_longer(clean_1,
  cols = starts_with("X"),
  names_to = "Year",
  values_to = "Value",
  values_transform = list(Value = as.character))
```

```
clean_3 <- mutate(clean_2, Year = str_remove(Year, "X"))
```

```
clean_4 <- clean_3 |> mutate(Year = as.numeric(Year), Value = as.numeric(Value))
```

```
clean_data <- clean_4 |>
  pivot_wider(names_from = DataSeries, values_from = Value)
singapore <- clean_data |>
  as_tsibble(index = Year)
singapore
```

```
## # A tsibble: 66 x 3 [1Y]
##   Year 'Total Fertility Rate (TFR)' 'Total Live-Births'
##   <dbl> <dbl> <dbl>
## 1 1960 5.76 61775
## 2 1961 5.41 59930
## 3 1962 5.21 58977
## 4 1963 5.16 59530
## 5 1964 4.97 58217
```

```
## 6 1965 4.66 55725
## 7 1966 4.46 54680
## 8 1967 3.91 50560
## 9 1968 3.53 47241
## 10 1969 3.22 44562
## # i 56 more rows
```

The data has been extracted, and the EDA is only on the Total Fertility Rate (TFR) and Total Live-Births (TLB).

```
singapore |> autoplot(`Total Fertility Rate (TFR)` ) +
  labs(title = "Total Fertility Rate (1960-2024)",
        caption = "Fig. 1 Plot of the Total Fertility Rate in Singapore from 1960 to 2024.",
        y = "Total Fertility Rate",
        x = "Year") +
  theme(plot.caption = element_text(hjust = 0.5),
        plot.title = element_text(hjust = 0.5))
```

```
## Warning: 'autoplot.tbl_ts()' was deprecated in fabletools 0.6.0.
## i Please use 'ggtime::autoplot.tbl_ts()' instead.
## i Graphics functions have been moved to the {ggtime} package. Please use
## 'library(ggtime)' instead.
## This warning is displayed once per session.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

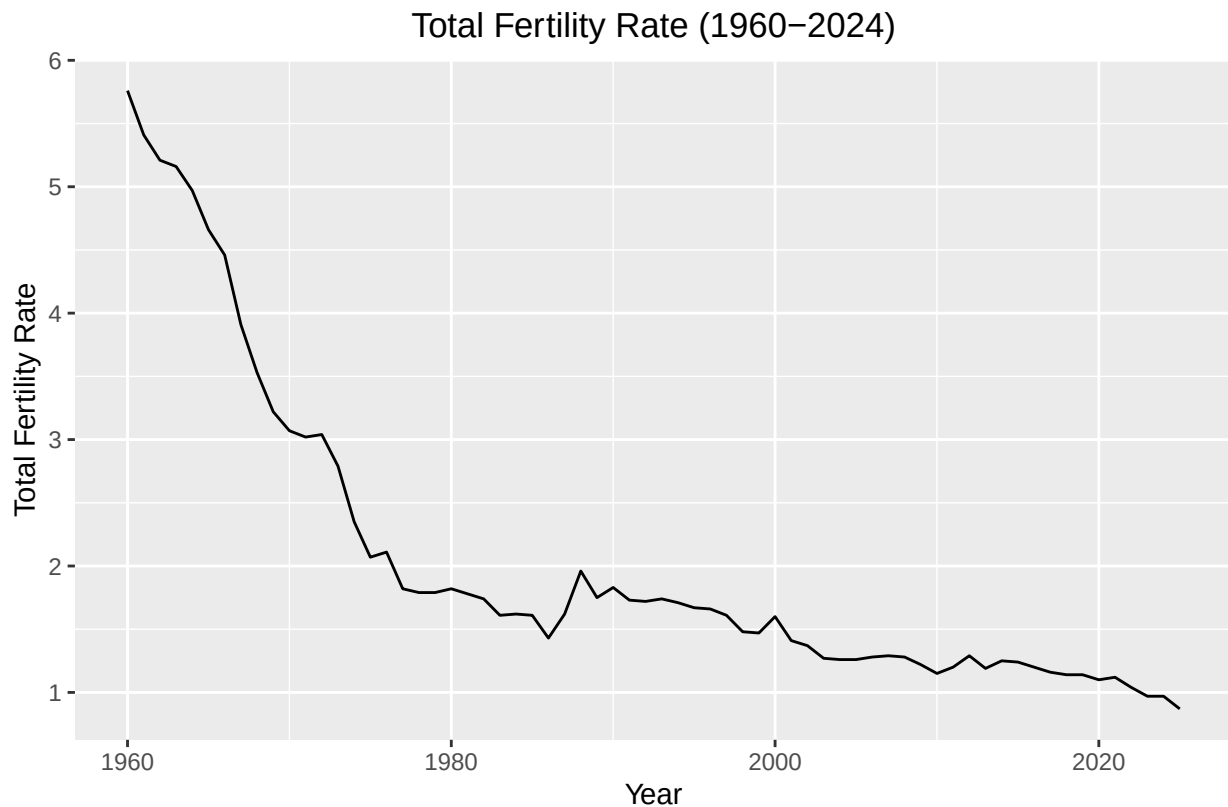


Fig. 1 Plot of the Total Fertility Rate in Singapore from 1960 to 2024.

```
singapore |> autoplot(`Total Live-Births`) +
  labs(title = "Total Live-Births (1960-2024)",
        caption = stringr::str_wrap("Fig. 2 Plot of the Total Live-Births in Singapore from 1960 to 2024"),
        y = "Total Live-Births",
        x = "Year") +
  theme(plot.caption = element_text(hjust = 0.5),
        plot.title = element_text(hjust = 0.5))
```

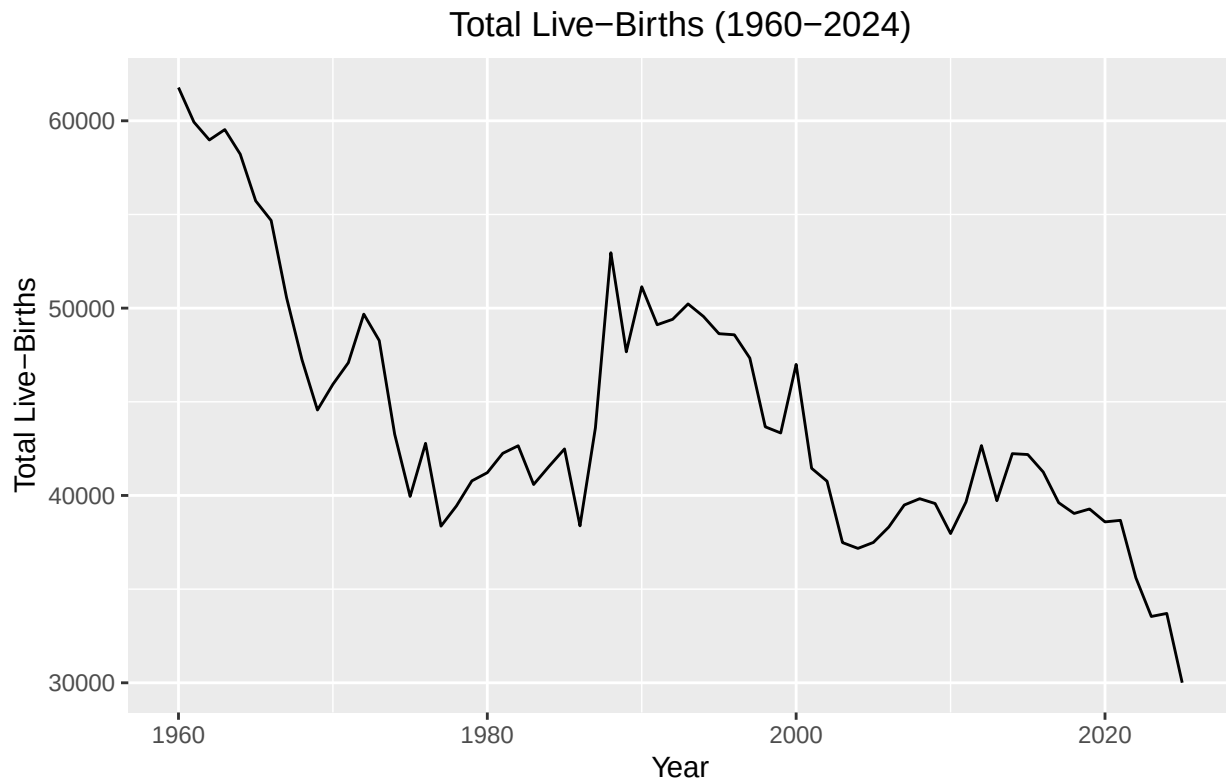


Fig. 2 Plot of the Total Live-Births in Singapore from 1960 to 2024 shows that it follows a roughly similar trend as the plot of the Total Fertility Rate.

While figure 1 and figure 2 look different we can see that they show spikes and drops at similar points. There is a spike around 1988 in the plot of Total Fertility Rate and a large spike around the same year in the plot of the Total Live-Births. This can also be seen with a spike around 1972, as well as drops around 1969, 1987, 2003, and 2024.

```
log_TFR <- log(singapore$`Total Fertility Rate (TFR)`)
log_TLB <- log(singapore$`Total Live-Births`)
```

```
singapore |> autoplot(log_TFR) +
  labs(title = "Log-Transformed Total Fertility Rate (1960-2024)",
        caption = stringr::str_wrap("Fig. 3 Plot of the Total Fertility Rate with a log transformation in"),
        y = "log(Total Fertility Rate)",
        x = "Year") +
  theme(plot.caption = element_text(hjust = 0.5),
        plot.title = element_text(hjust = 0.5))
```

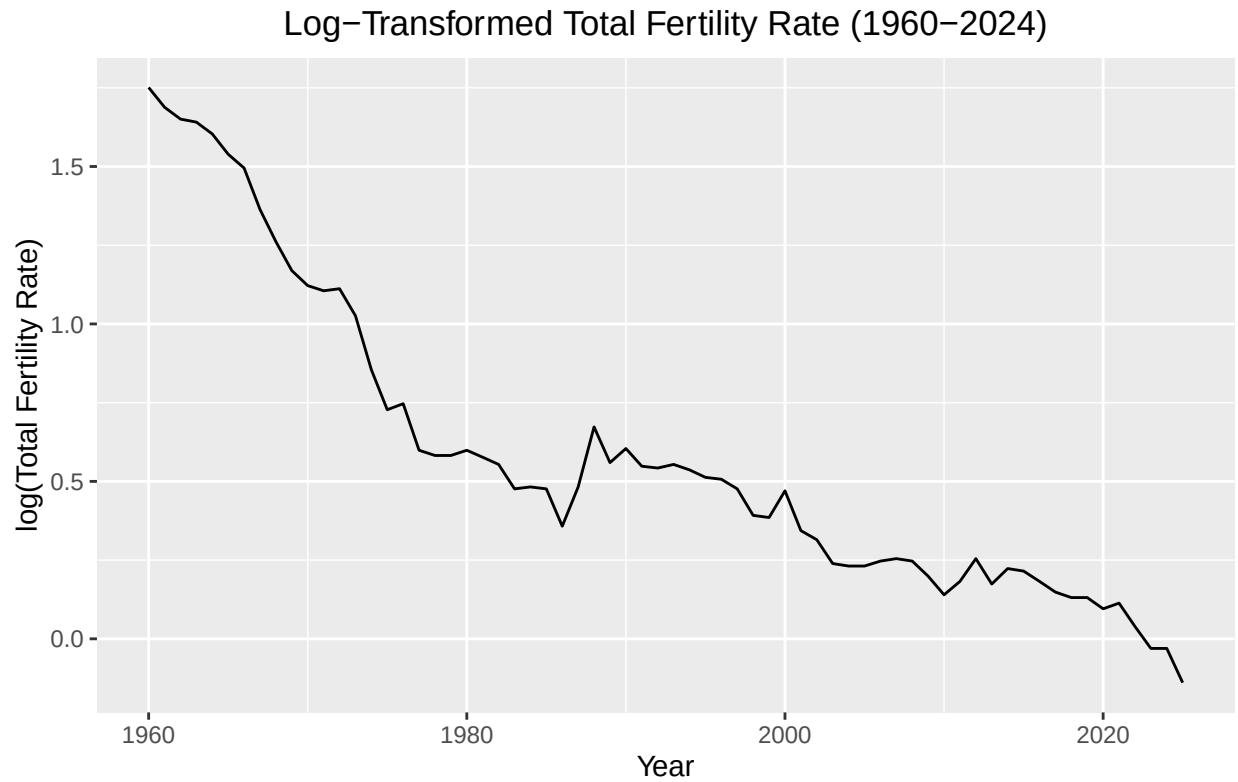


Fig. 3 Plot of the Total Fertility Rate with a log transformation in Singapore from 1960 to 2024.

```
singapore |> autoplot(log_TLB) +
  labs(title = "Log-Transformed Total Live-Births (1960-2024)",
        caption = stringr::str_wrap("Fig. 4 Plot of the Total Live-Births with a log transformation in S",
        y = "log(Total Live-Births)",
        x = "Year") +
  theme(plot.caption = element_text(hjust = 0.5),
        plot.title = element_text(hjust = 0.5))
```

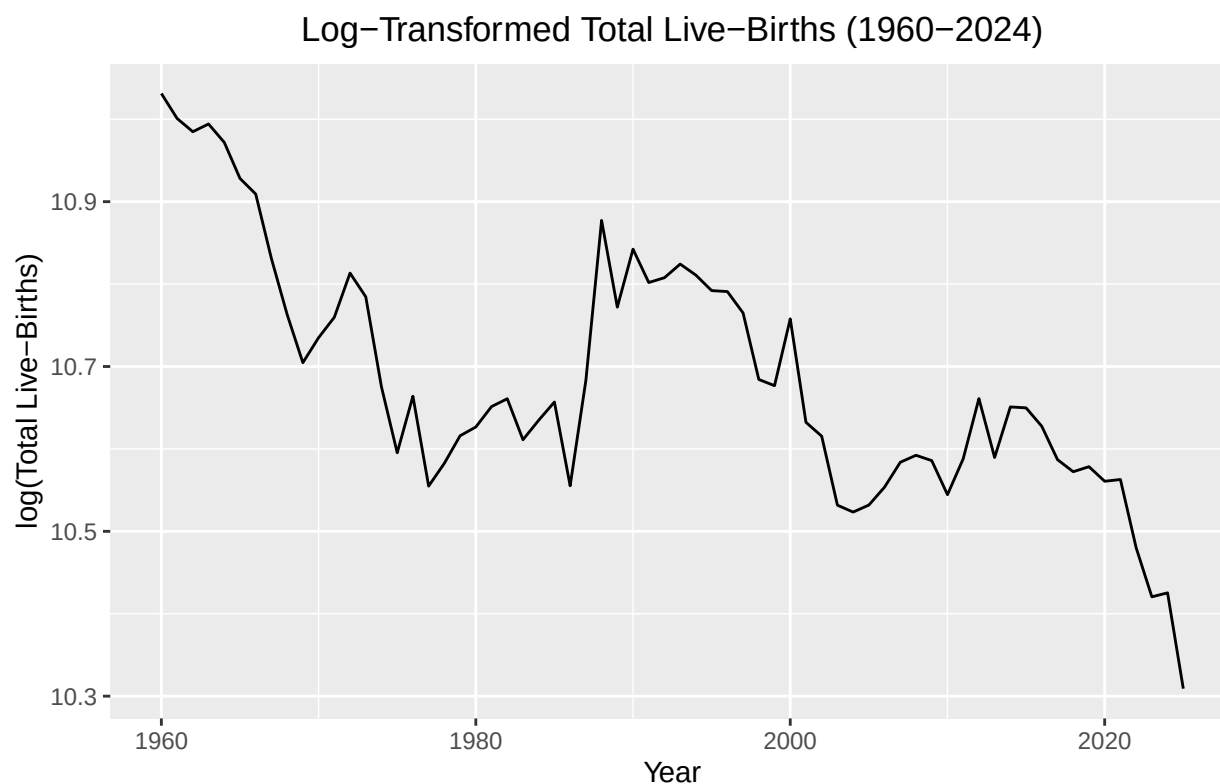


Fig. 4 Plot of the Total Live-Births with a log transformation in Singapore from 1960 to 2024.

```
singapore |> features(log_TFR, unitroot_kpss)
```

```
## # A tibble: 1 x 2
##   kpss_stat kpss_pvalue
##   <dbl>    <dbl>
## 1     1.49      0.01
```

Table 1. KPSS test of the log transformed Total Fertility rate in Singapore from 1960-1924. The p-value is 0.01, which is significant at the $\alpha = 0.05$ significance level.

```
singapore |> features(log_TLB, unitroot_kpss)
```

```
## # A tibble: 1 x 2
##   kpss_stat kpss_pvalue
##   <dbl>    <dbl>
## 1     0.996      0.01
```

Table 2. KPSS test of the log transformed Total Live-Birth rate in Singapore from 1960-1924. The p-value is 0.01, which is significant at the $\alpha = 0.05$ significance level.

Figures 3 and 4, along with the p-values from tables 1 and 2 both being significant at the 0.05 significance level, show that a log transformation seems to help a bit but they both still need to be stabilized more. To do this, they will be differenced.

```
TFR <- difference(log(singapore$`Total Fertility Rate (TFR)`), 1)
```

```
TLB <- difference(log(singapore$`Total Live-Births`), 1)
```

```
singapore |> autoplot(TFR) +
  labs(title = "First-Order difference of Log TFR (1960-2024)",
        caption = stringr::str_wrap("Fig. 5 Plot of the first-order difference of the log transformed Total Fertility Rate in Singapore from 1960 to 2024."),
        y = "Change in log(TFR)",
        x = "Year") +
  theme(plot.caption = element_text(hjust = 0.5),
        plot.title = element_text(hjust = 0.5))
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_line()').
```

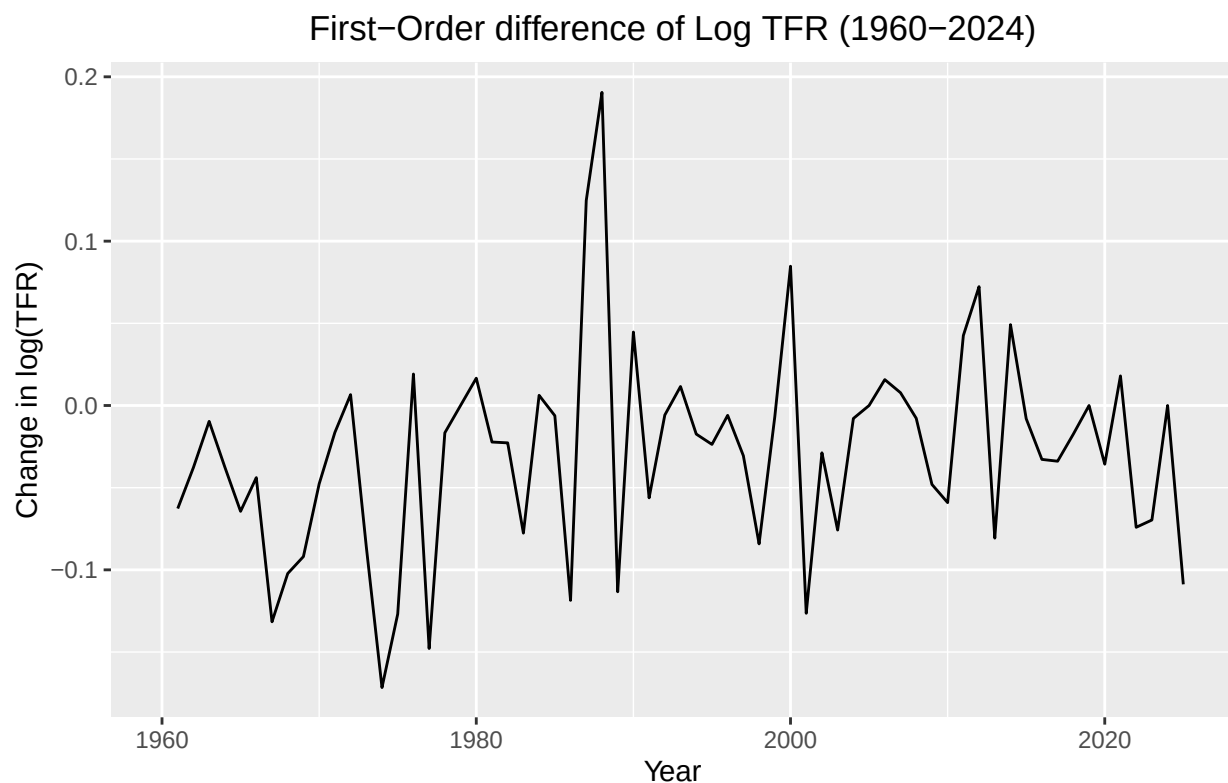


Fig. 5 Plot of the first-order difference of the log transformed Total Fertility Rate in Singapore from 1960 to 2024.

```
singapore |> autoplot(TLB) +
  labs(title = "First-Order difference of Log TLB (1960-2024)",
        caption = stringr::str_wrap("Fig. 6 Plot of the first-order difference of the log transformed Total Live-Births in Singapore from 1960 to 2024."),
        y = "Change of log(TLB)",
        x = "Year") +
  theme(plot.caption = element_text(hjust = 0.5),
        plot.title = element_text(hjust = 0.5))
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_line()').
```

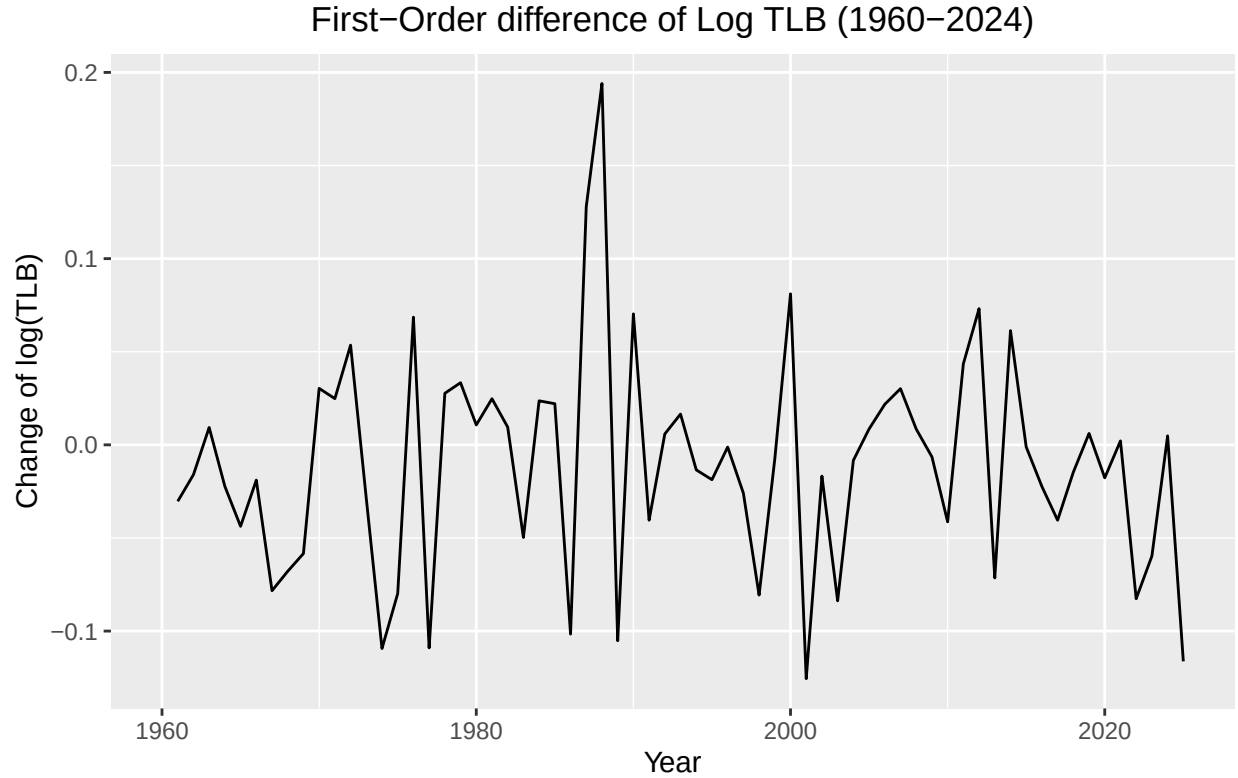


Fig. 6 Plot of the first-order difference of the log transformed Total Live-Births in Singapore from 1960 to 2024.

Plots 5 and 6 no longer show a downward trend with a first-order difference of the log transform of each variable.

```
singapore |> features(TFR, unitroot_kpss)
```

```
## # A tibble: 1 x 2
##   kpss_stat kpss_pvalue
##   <dbl>      <dbl>
## 1      0.370      0.0902
```

Table 3. KPSS test of the first-order differenced log transformed Total Fertility rate in Singapore from 1960-1924. The p-value is 0.09, which is not significant at the $\alpha = 0.05$ significance level.

```
singapore |> features(TLB, unitroot_kpss)
```

```
## # A tibble: 1 x 2
##   kpss_stat kpss_pvalue
##   <dbl>      <dbl>
## 1      0.114      0.1
```

Table 4. KPSS test of the first-order differenced log transformed Total Live-Birth rate in Singapore from 1960-1924. The p-value is 0.1, which is not significant at the $\alpha = 0.05$ significance level.

Tables 3 and 4 confirm what figures 5 and 6 showed, when both Total Live-Births and Total Fertility rate are log transformed and then differenced, they are stationary. In a KPSS test, if the p-value is less than 0.05,

the null-hypothesis that the series is stationary, is rejected. However, if the p-value is greater than 0.05, the null-hypothesis is accepted and the series is found to be stationary.

```
singapore |> ACF(TFR) |>
  autoplot() +
  labs(title = "ACF of First-Order Differenced Log TFR (1960-2024)",
        caption = stringr::str_wrap("Fig. 7 Autocorrelation Function of the stationary Total Fertility Rate"),
        theme(plot.caption = element_text(hjust = 0.5),
              plot.title = element_text(hjust = 0.5))
)
```

```
## Warning: 'autoplot.tbl_cf()' was deprecated in feasts 0.4.2.
## i Please use 'ggtime::autoplot.tbl_cf()' instead.
## i Graphics functions have been moved to the {ggtime} package. Please use
##   'library(ggtime)' instead.
## This warning is displayed once per session.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

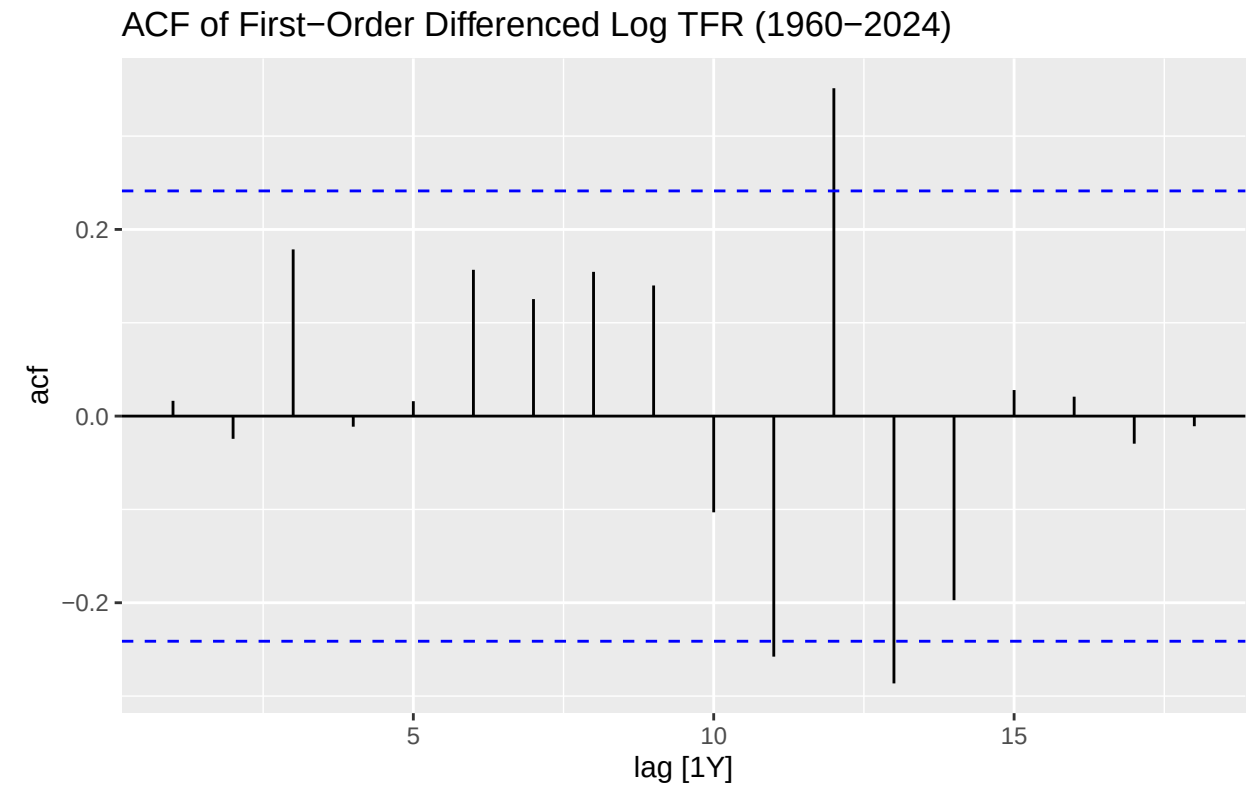


Fig. 7 Autocorrelation Function of the stationary Total Fertility Rate data in Singapore from 1960-2024.

```
singapore |> ACF(TLB) |>
  autoplot() +
  labs(title = "ACF of First-Order Differenced Log TLB (1960-2024)",
        caption = stringr::str_wrap("Fig. 8 Autocorrelation Function of the stationary Total Live-Births"),
        theme(plot.caption = element_text(hjust = 0.5),
              plot.title = element_text(hjust = 0.5)))
```

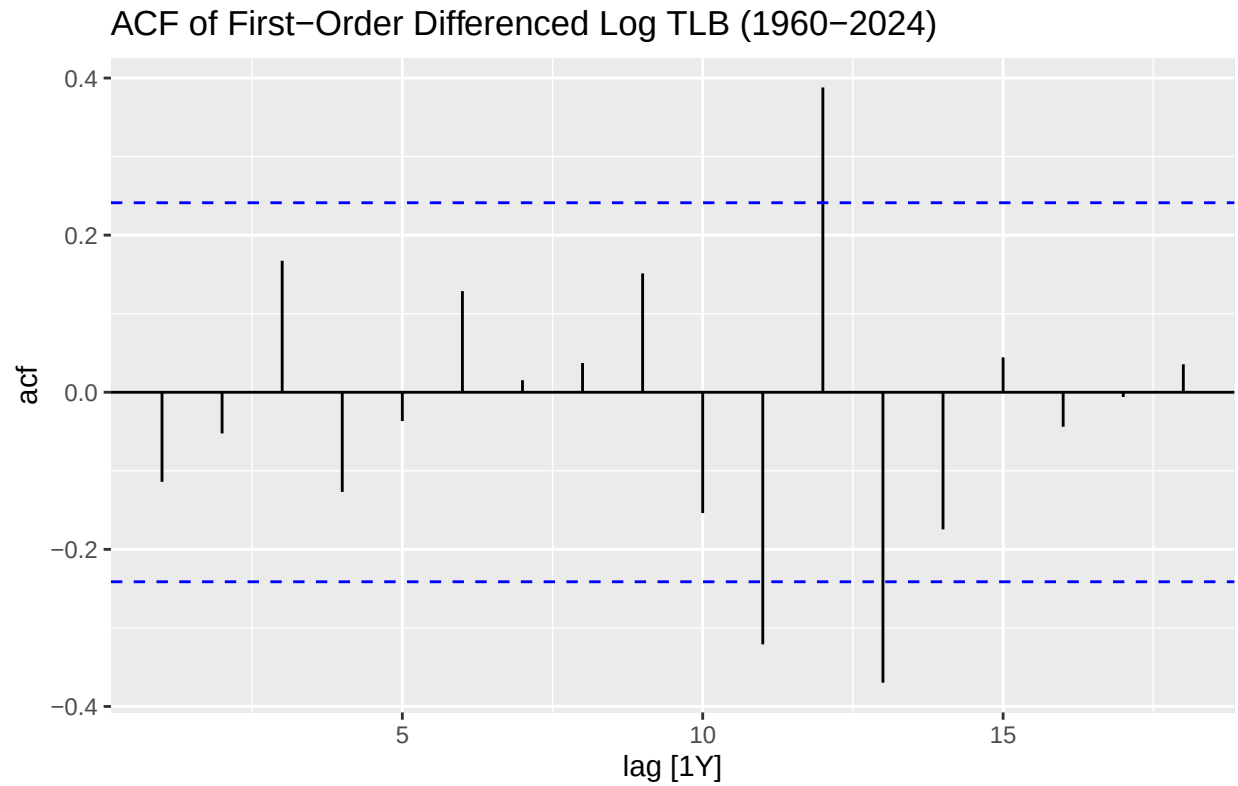



Fig. 8 Autocorrelation Function of the stationary Total Live-Births data in Singapore from 1960–2024.

```
singapore |> PACF(TFR) |>
  autoplot() +
  labs(title = "PACF of First-Order Differenced Log TFR (1960-2024)",
        caption = stringr::str_wrap("Fig. 9 Partial Autocorrelation Function of the stationary Total Fer",
        theme(plot.caption = element_text(hjust = 0.5),
              plot.title = element_text(hjust = 0.5))
  )
```

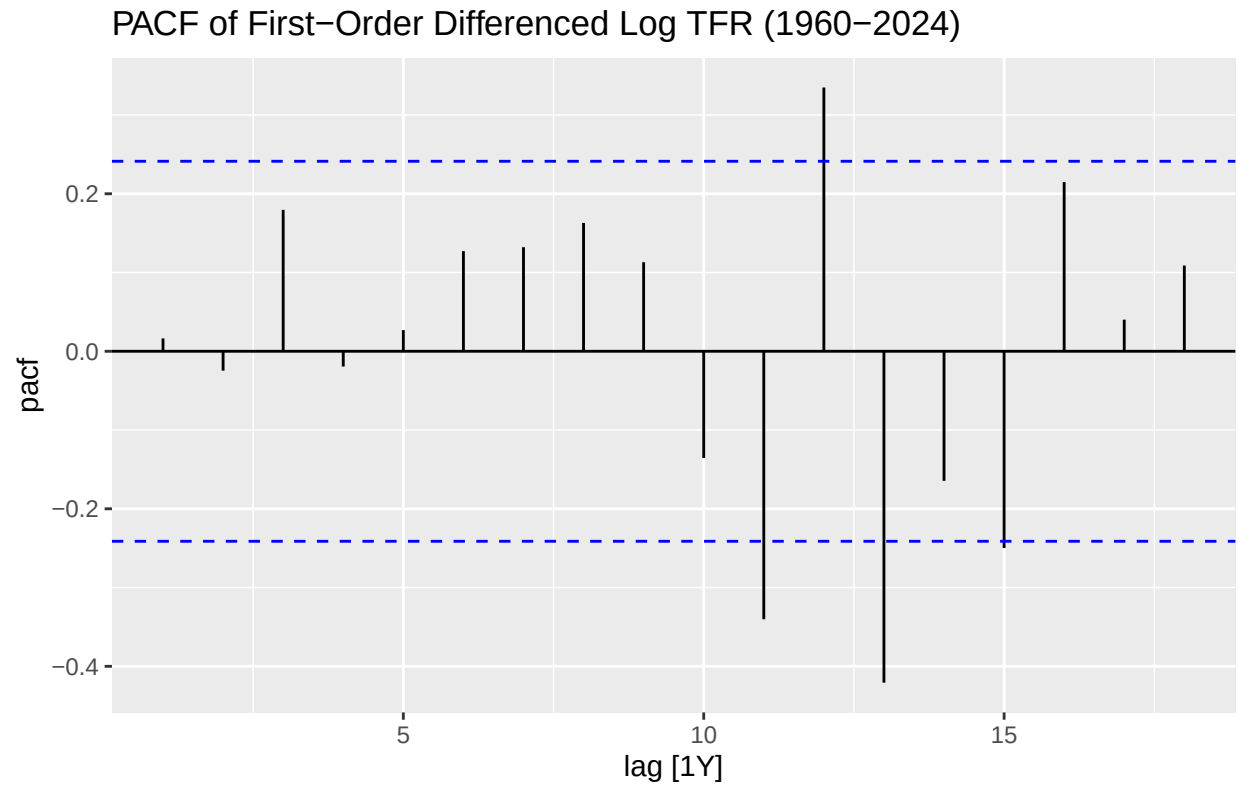


Fig. 9 Partial Autocorrelation Function of the stationary Total Fertility Rate data in Singapore from 1960–2024.

```
singapore |> PACF(TLB) |>
  autoplot() +
  labs(title = "PACF of First-Order Differenced Log TLB (1960-2024)",
        caption = stringr::str_wrap("Fig. 10 Partial Autocorrelation Function of the stationary Total Li",
        theme(plot.caption = element_text(hjust = 0.5),
              plot.title = element_text(hjust = 0.5)))
```

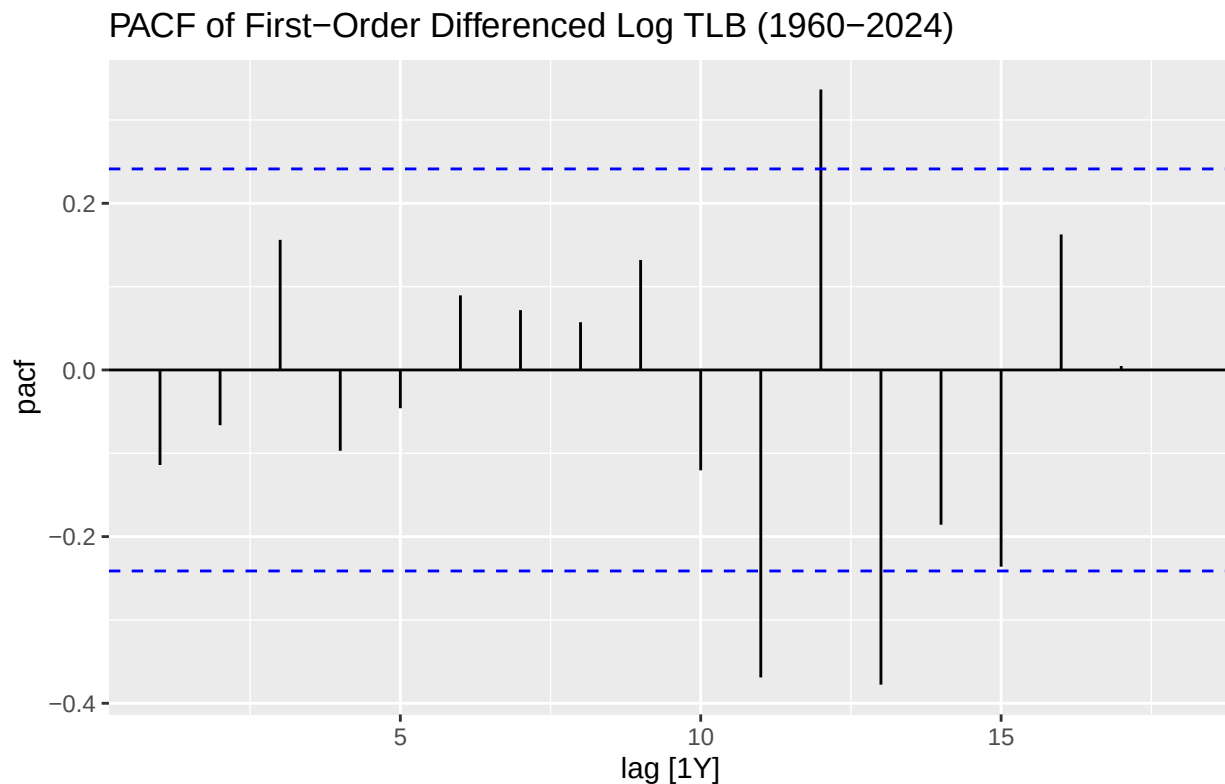


Fig. 10 Partial Autocorrelation Function of the stationary Total Live-Births data in Singapore from 1960–2024.

The spikes at lags 11,12,13 in figures 7,8,9, and 10 show that there’s a cyclic behaviour. The year of the Dragon historically has had an effect on birth rates in Singapore. Culturally, it is “considered auspicious for births” (The Straits Times). The Year of the Dragon comes every 12 years, which lines up perfectly with the spikes.

```
TFR_Model1 <- singapore |>
  model(
    model1_baseline = ARIMA(log_TFR ~ pdq(0,1,0)),
    model1_auto = ARIMA(log_TFR))
TFR_Model1
```

```
## # A mable: 1 x 2
##       model1_baseline      model1_auto
##       <model>             <model>
## 1 <ARIMA(0,1,0) w/ drift> <ARIMA(0,1,0) w/ drift>
```